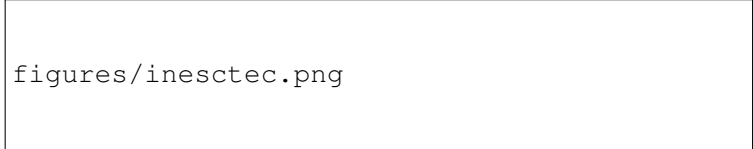


RicoSLAM: Localization and Mapping in Dynamic Environments

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Resumo

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Abstract

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Next, a special thanks to my colleagues from CRIIS, including from the laboratories iilab – industry and innovation lab and TRIBElab – Laboratory of Robotics and IoT for Smart Precision Agriculture and Forestry. Although my research is related to the intralogistics industry and more related to iilab’s scope and mission, I began my PhD journey working side by side with TRIBElab, as the lab is located on the campus of the Faculty of Engineering, University of Porto (FEUP). This experience was excellent in terms of the team environment and spirit, while also providing the opportunity to interact daily with my mates from the master’s degree, namely André and Daniel. Additionally, the TRIBElab leader, Filipe Santos, was one of the most important people who helped me decide that staying at INESC TEC would be the best choice to continue my journey and pursue a PhD degree, given the interaction with real-world applications and the opportunity to move beyond simulators and the theoretical world. So, a special thanks to him for that support, and to André and Daniel, two close companions and friends who supported me greatly throughout my PhD. After I moved to the iilab facilities, it has been a fantastic journey, marked by the creation of new team partnerships and demonstrating that working side by side with colleagues from the same application field and different research areas can result in fascinating research experiences. A special thanks to my team leader Héber, also the co-supervisor of my PhD thesis, Prof. Manuel Silva, for his guidance on the GreenAuto project directly linked with my PhD research and all the support given by him and the current and past CRIIS coordination team, to all my teammates of the Mobile Robotics Development Team (MRDT), and all my colleagues from iilab that I interact daily with and has been a pleasure to work with them.

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Ricardo B. Sousa

*“I don’t stop when I’m tired.
I stop when I’m done.”*

David Goggins

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Abbreviations

ADRA AI, Data and Robotics Association. 2

AI Artificial Intelligence. 2

AMR Autonomous Mobile Robot. 1–4

BDVA Big Data Value Association. 2

CAGR Compound Annual Growth Rate. 1

CLAIRE Confederation of Laboratories for Artificial Intelligence Research in Europe. 2

CRIIS Centre for Robotics in Industry and Intelligent Systems. 6

EKF Extended Kalman Filter. 8

ELLIS European Laboratory for Learning and Intelligent Systems. 2

EU European Union. 2

EurAI European Association for Artificial Intelligence. 2

FEUP Faculty of Engineering, University of Porto. 6

FOV Field-of-View. 4

ICP Iterative Closest Point. 5

iilab industry and innovation lab. 6

INESC TEC Institute for Systems and Computer Engineering, Technology and Science. 6

PPS Product, Processes, or Services. 3, 6

ROS Robot Operating System. 8

SDG Sustainable Development Goal. 6, 7

SLAM Simultaneous Localization and Mapping. 1, 2, 4–9

SRIDA Strategic Research, Innovation, and Deployment Agenda. 2, 3

UN United Nations. 7

w.r.t. with respect to. 6

WP Work Package. 3, 6

Chapter 1

Introduction

This research, *RicoSLAM: localization and mapping in dynamic environments*, aims to advance the current state of the art in Simultaneous Localization and Mapping (SLAM). This work focuses on improving localization and mapping performance when performing SLAM in dynamic environments, using wheeled odometry and 2D laser scanner data from sensors onboard industrial ground mobile robots. SLAM in dynamic environments is closely tied to lifelong SLAM and long-term operation of mobile robots, enabling robots to maintain accurate localization and mapping over extended periods while adapting to continuous environmental changes.

This chapter is divided into seven sections. First, Section 1.1 contextualizes this research work and describes the primary challenges related to localization and mapping in dynamic environments. Section 1.2 outlines the key motivations and reasons for undertaking this research. Next, Section 1.3 introduces the problem that this PhD Thesis seeks to address. Section 1.4 presents the main research questions and hypotheses that guided the research work. Then, Section 1.5 defines the scope and objectives of the PhD Thesis. Section 1.6 highlights the unique contributions the research makes to the literature, emphasizing its novelty and impact. Lastly, Section 1.7 outlines the document's organization. A final note is provided regarding the funding that supported the work carried out throughout the PhD Thesis.

1.1 Context

Nowadays, Autonomous Mobile Robots (AMRs) are being increasingly deployed in several industries, including logistics, retail, warehousing, automotive and aerospace manufacturing, agriculture, and healthcare. AMRs may automate tasks such as material transport, automated inspection, and security monitoring. By improving operational efficiency, flexibility and productivity, AMRs help companies achieve higher efficiencies and reduce production costs. Consequently, these improvements contribute to the growth of the global economy and increase the worldwide interest in the AMR market. According to a study by ABI Research, the global robotics market is estimated to be roughly US\$50 billion in 2025, growing at a Compound Annual Growth Rate (CAGR) of about 14% to reach approximately US\$111 billion by 2030. This market growth is driven by labour workforce shortages, supply chain reformation, and high turnover rates, with the concurrent increasing consumer demand. Mobile robots in particular can mitigate these challenges by expediting the loading/unloading of trucks, automating inventory management, and performing last-mile delivery services [1].

An AMR must interpret its surroundings by extracting meaningful data from its sensors to localize itself within the environment, enabling autonomous navigation [2]. SLAM addresses this perception and localization

challenge by concurrently incorporating the robot state estimation into the mapping process to build a representation of the environment as perceived by the robot. The state is typically described by the robot's pose (position and orientation), although other variables, such as velocity and calibration parameters, may also be included in the state. As for the environment representation, the mapping process involves creating a map incrementally, built as the robot explores unknown areas or refines it through passages in known locations, to describe the environment in which the robot operates. The map representation serves multiple purposes, such as path planning, providing a visual reference for human operators, while allowing the robot to reset its localization error when revisiting known areas [3].

In a static scene, the robot would only need to map once because the map and sensor data would remain consistent with the environment throughout time. However, industrial and intralogistics locations, agricultural sites, warehouses, and service-oriented applications, such as shopping centers and healthcare institutions, have inherently moving elements in the scene (humans, objects), reconfiguration of the environment (reorganization of machines, patient beds), and appearance variations (lighting, weather, seasonal, or daytime changes). The deployment of AMRs in these varying conditions are a challenge for the SLAM system, where the system should decide how (consider the most current state, the evolution of the environment dynamics, or only the most static changes?) and when to update the map (when the variations occur, or only after a certain time?). This challenge is also known as the stability-plasticity dilemma, where a lifelong SLAM system should both adapt to new environment changes and preserve old states over time [4]. Furthermore, the map would grow indefinitely when gathering new information from the environment without any map management system. This ever-growing problem poses another challenge for the SLAM system in the long-term due to the limited computational resources of the mapping vehicle. Indeed, the map size should be dependent on the environment area and not on the trajectory length of the mapping process nor operation time, only growing when the robot would explore unknown locations [5].

[TODO: Brief literature analysis (only after finishing Literature Review chapter)]

1.2 Motivation

In May 2021, a joint partnership between major European organizations – Big Data Value Association (BDVA), Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE), European Laboratory for Learning and Intelligent Systems (ELLIS), European Association for Artificial Intelligence (EurAI), and euRobotics – founded the AI, Data and Robotics Association (ADRA), formed from the European Partnerships in Cluster 4 (digital, industry, and space) in Horizon Europe. Since then, ADRA, in collaboration with the European Commission and other community members, has worked on the Strategic Research, Innovation, and Deployment Agenda (SRIDA) [6]. The most recent version, published in February 2024, reflects on European strategies and goals that Artificial Intelligence (AI), data, and robotics can address on global challenges. This version outlines the ADRA's Vision 2030, the global challenges and primary strategic objectives that the ADRA partnership seeks to address, major trends and gaps, as well as recommendations for the European Union (EU)'s 2025–2027 strategic plan in alignment with the SRIDA.

SRIDA [6] identifies the need for a new wave of AI, data, and robotics methods that can tackle complex societal and industrial challenges. Driven by the accelerating progress of AMRs, these challenges could lead to the development of self-supervised robot perception and autonomous control mechanisms, enabling predictive and safe robot operation without human intervention. Moreover, SRIDA acknowledges that autonomy in

robotics is currently constrained by short-term, localized environments with controlled uncertainty. There is an urgent need for higher levels of autonomy, particularly in complex, unknown, and dynamic environments. Indeed, these environments often include unanticipated scenarios, which robots must be capable of handling and adapting to. Consequently, robots may present significant risks to both people and the environment, underscoring the need for reliable and trustworthy autonomous operations. As a result, SRIDA motivates research on improving localization and mapping processes for industrial mobile robots within dynamic environments, aiming to increase their levels of autonomy. This expansion of the range of scenarios where robots can operate without human intervention contributes to the long-term operation and reliable, lifelong autonomy of AMRs.

Furthermore, the logistics industry has high dynamic environments with constantly shifting elements, such as pallets, boxes, workers, moving AMRs (base platforms, rack and cart lifting robots, car pulling AMRs, moving pallet platforms, conveyor robots, mobile manipulators, pallet jack AMRs, tugger robots to tow carts like a convoy, and autonomous forklifts [7]), and manually operated mobile machines. Additionally, logistics warehouse environments may have featureless characteristics that make it difficult for AMRs to localize within the environment over time. One situation that is particularly critical to address in dynamic logistics environments involving robots' localization is cases involving random slotting warehouse strategies, which optimize space and enhance efficiency in modern operations. In random slotting, products are stored in any available space in the warehouse based on their characteristics (weight and size), thereby reducing replenishment time and increasing warehouse efficiency [8]. Subsequently, the interpretation and processing of dynamic changes in the environment by AMRs would enable the maintenance of an up-to-date map representation of the robots' surroundings in logistics environments. An up-to-date representation would allow the use of dynamic elements in the scene that are temporarily static, thereby improving the robot localization estimation in the environment.

Finally, the Work Package (WP) 10 of the *GreenAuto – Green Innovation for the Automotive Industry* [9] project aims to develop technologies that reduce the production costs of electric vehicles, which are currently higher than those of traditional internal combustion engine alternatives. The use of autonomous logistic platforms on the factory floor, which either directly or indirectly assist in the production process, is already well established. However, there is significant room for improvement, particularly in extending the applicability of those systems to less structured environments. WP10 proposes the development of a navigation system for AMRs that is robust in complex environments. One of the key deliverables from WP10 is the Product, Processes, or Services (PPS) 18, a navigation system for mobile robots designed to be both robust and efficient in automotive shop floor settings. Consequently, PPS18 motivates the research and development of localization and mapping algorithms that account for dynamic elements in the environment, aiming to enhance the robustness of the final autonomous system.

1.3 Problem Statement

As discussed in Section 1.1, a map representation in static environments does not require updates, as it remains consistent with the surrounding environment that has been explored. However, most use cases for AMRs involve dynamic environments with moving elements or reconfigurations that affect the consistency of the map over time. Consequently, the system must determine how and when to update the map, a challenge known as the stability-plasticity dilemma. In this context, a long-term SLAM system must adapt to new changes while preserving older features, particularly when the environment reverts to previous states [4].

This PhD thesis addresses the problem of localizing a mobile robot within dynamic indoor environments. This problem is directly related to SLAM, given the need to update the map concurrently with localization upon dynamic changes in the robot’s surroundings. Furthermore, this research is not limited to a specific type of dynamics, for example, only detecting semi-static dynamics – changes between running sessions, *i.e.*, changes occurring outside of the sensors’ Field-of-View (FOV) at different time instants that, when observed again by the sensors, appear static but elements were effectively moved in the scene – or only dealing with highly dynamic elements (*e.g.*, moving persons or objects). Instead, this thesis studies the detection of dynamic changes by comparing the current observation with a local reference map representation.

In terms of the sensor setup, this PhD thesis assumes that the robot is equipped with at least one 2D laser scanner and, optionally, wheeled odometry data. This sensor setup is commonly found in AMRs within the logistics industry, where wheeled odometry is typically estimated by counting the ticks from the motor encoders while taking into account the robot’s kinematic model [10]. In addition, AMRs often use 2D laser scanners not only for navigation but also as safety devices, in compliance with the ISO 10218-1:2025 [11] and ANSI/RIA R15.06-2025 [12] standards, which ensure a safe stopping distance and help prevent collisions with obstacles. Consequently, this thesis focuses solely on range data.

1.4 Research Questions and Hypotheses

In response to the problem outlined in Section 1.3, this thesis proposes some research questions aimed at addressing and solving the identified problem. For each of these research questions, we formulate hypotheses that guided the investigation toward feasible solutions.

Research Question 1. *How to improve the mapping results when performing SLAM with a ground mobile robot in a dynamic indoor environment?*

In dynamic environments, SLAM algorithms can be affected by the presence of moving objects and changing surroundings. One key contributing factor to reduced SLAM accuracy is the inability of traditional static map representations to stay updated with real-time changes in the environment, leading to mismatches between current sensor observations and previously mapped data. As a result, the correspondence between these observations and the reference environment representation becomes weaker, leading to degraded map quality. Another issue arises from the growing amount of information to process over time. As the robot moves, the accumulation of data from sensors can lead to redundant information, especially in environments where sensor noise, matching errors, or outlier observations occur frequently. Consequently, the appearance of “fat” walls in the map representation obtained from SLAM can occur. Another consequence of this issue is that, while the map may appear fine locally, distortions and inaccuracies can accumulate at the global level, leading to inconsistencies between the map representation and the real environment itself.

Given the previous challenges, we propose research hypotheses aimed at improving mapping results during SLAM in dynamic indoor environments. The primary focus is on removing outdated or redundant observations from the environment representation to reduce the noise introduced by both the sensor and the matching process, continuously updating the map with the latest sensor observations to increase the inliers matching ratio, while exploring concurrently offline global bundle adjustment techniques to remove distortions from the global map.

Hypothesis 1.1. *Exploring the application of ray tracing techniques to update map representations based on the latest sensor observations continuously. These techniques can help determine whether a sensor observation*

corresponds to a new point or an update to an outdated one, depending on whether the latest ranging data is closer than the existing map point or if the sensor observation is passing through an existing feature.

Hypothesis 1.2. *Researching techniques for robot relocalization on existing graphs representations, avoiding the addition of redundant data to the map. Linking graph growth to the spatial extent of the environment explored, this approach could help avoid unnecessary graph expansion, keeping the graph size manageable and improving computational efficiency without the graph size being dependent on the operation time.*

Hypothesis 1.3. *Investigating the offline removal of dynamic points using ray tracing techniques during the generation of the environment representation. This representation could be leveraged for more accurate mobile robot localization by excluding transient or moving objects from the map.*

Hypothesis 1.4. *Studying offline global map refinement techniques, removing dynamic points in the process, while trying to remove map distortions from the online-based map representation.*

Research Question 2. *How to integrate environment dynamics in the localization process when performing SLAM with a ground mobile robot in a dynamic indoor environment?*

The localization process in SLAM is also dependent on the quality of the sensor data, especially when dynamic elements are present in the environment. Moving objects, such as people, vehicles, or machinery, can introduce significant challenges during the data registration step. When these dynamic observations are considered during data registration without any outliers rejection technique, could introduce errors on pose estimation, potentially causing the robot's localization to drift or even fail entirely. In order to mitigate the effects of environment dynamics in the localization process, we explore several strategies formulated by the following research hypotheses. These hypotheses focus on extracting geometric features from the 2D laser scanner data, using robust kernels during the data registration process to reduce the influence of outliers while applying distance maps to avoid the traditional association step to speed up the data registration process. Additionally, implementing local map strategies to continuously update the local environment representation with the latest sensor observations can help maintain a more accurate knowledge of the robot's surroundings, improving localization stability in highly dynamic environments.

Hypothesis 2.1. *Exploring the use of geometric feature extraction, such as corners or lines, from 2D laser scanner data for localization purposes, rather than relying solely on raw ranging data. This approach may help filter out irrelevant data, providing more meaningful and stable features for localization, even in dynamic environments.*

Hypothesis 2.2. *Researching the use of distance map data structures for 2D point cloud registration. Distance maps can provide access to the nearest-neighbor point distance, enabling an Iterative Closest Point (ICP) [13]-like registration process without the need for the traditional association step. By applying robust kernels, this approach can improve the handling of dynamic environment point observations by considering them as outliers in the alignment optimization process or assigning them a lower weight.*

Hypothesis 2.3. *Exploring local map strategies to continuously update the local environment representation with the latest sensor observations. Unlike scan-based localization, which may be more prone to false positives or negatives, local maps facilitate loop closure detection through a scan-to-map alignment rather than scan-to-scan approaches that consider singular scans. Moreover, local maps can enhance the localization stability of*

mobile robots in highly dynamic environments, as they are continuously updated locally with the latest sensor observations, while allowing the incorporation of ray-tracing-based techniques to detect dynamic elements and update the local representation accordingly.

1.5 Scope and Objectives

Given the problem statement in Section 1.3, alongside the research questions outlined in Section 1.4, and considering the current advancements and state of the art in localization and mapping in dynamic environments as discussed in Chapter 2, this research thesis aims to explore and advance developments in SLAM using wheeled odometry and 2D laser scanner data, particularly focused on indoor industrial environments. We assume that the plant floor is planar, allowing just a 2D pose estimation to localize the robot in the environment.

In order to achieve the thesis's main goal, the following efforts were undertaken:

- modifications to existing mobile platforms to enable data acquisition for the purposes of this PhD thesis;
- development of a line fitting-based corner-like feature extraction algorithm to capture corners geometric information from the environment on 2D laser scanners data;
- 2D point cloud data registration algorithms based on unsigned distance map data structures, proposing both point-to-point and point-to-plane error metrics;
- a full distance map-based graph SLAM framework, where the front-end estimates the robot pose with respect to (w.r.t.) a local reference representation (*e.g.*, singular scan or a local map), generating odometry and loop closure constraints, while the back-end of the framework implements a pose graph optimization using the `srrg2_solver` [14];
- ray tracing-based techniques to deal with dynamic observations on the 2D point cloud map representation of the environment;
- offline global bundle adjustment formulation based on distance maps to try to remove distortions from the global map.

Furthermore, while the degree-granting institution is the Faculty of Engineering, University of Porto (FEUP), the host institution of this PhD thesis is INESC TEC – Institute for Systems and Computer Engineering, Technology and Science. Specifically, all work conducted in this thesis was performed at iilab – industry and innovation lab, within CRIIS – Centre for Robotics in Industry and Intelligent Systems from INESC TEC. Throughout the PhD journey, the research was executed alongside PPS18 – *3D navigation system for mobile robotic equipment*, part of WP10 – *automated logistics for the automotive industry*, within the *GreenAuto Agenda: Green Innovation for the Automotive Industry* [9], as a contribution for the long-term SLAM research topic. This PPS involves several partners, including FEUP, INESC TEC, and the leading partner, Flowbotic Mobile Systems [15], responsible for the development of an autonomous forklift for the automotive industry.

Within the scope of the 2030 Agenda for sustainable development (see Figure 1.1), agreed by 195 nations in September 2015 [16], this research work falls under the following Sustainable Development Goals (SDGs):

- SDG8: Decent work and economic growth;
- SDG9: Industry, innovation and infrastructure;
- SDG12: Responsible Consumption and Production.

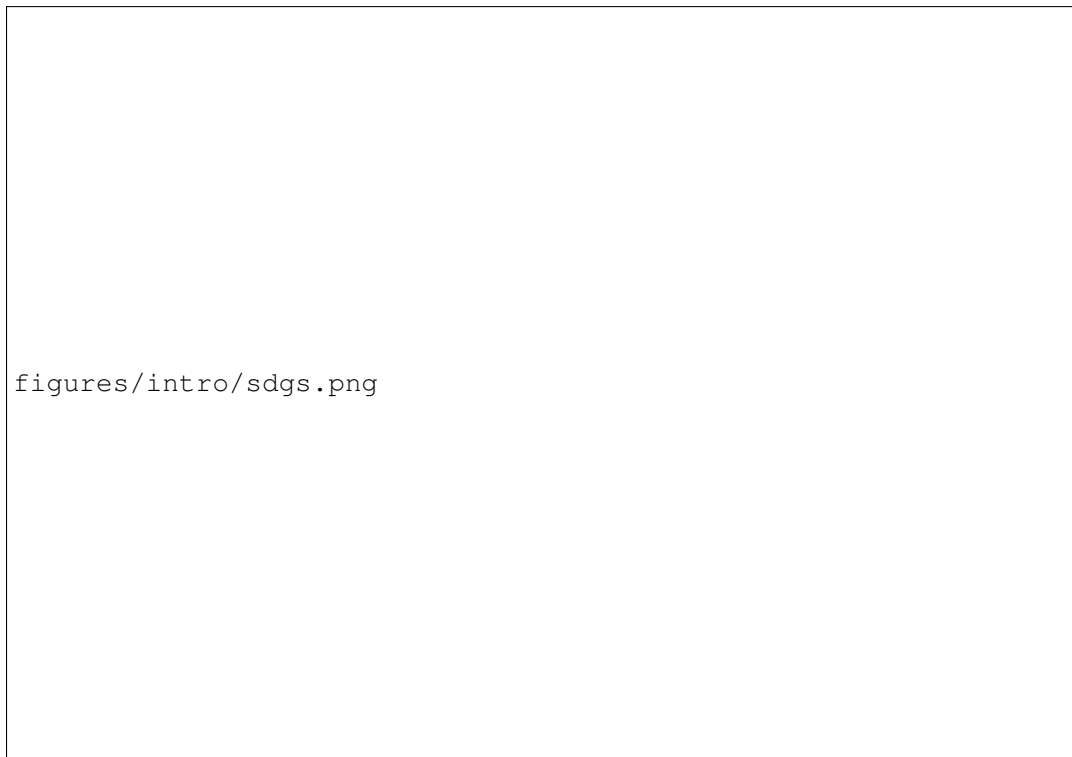


Figure 1.1: The United Nations (UN) Sustainable Development Goals (SDGs)

1.6 Contributions

This doctoral research advances localization and mapping for mobile robots in dynamic industrial environments by combining state of the art research, experimental evaluation, and system-level integration. The work resulted in peer-reviewed publications covering perception and feature extraction from 2D laser scanners, multimodal sensing integration, localization in dynamic environments, and SLAM benchmarking. Beyond the scientific contributions, this thesis also includes the public release of a dataset to support reproducibility and benchmarking, as well as the development of open-source software.

1.6.1 Main contributions

- [17] R.B. Sousa, H.M. Sobreira, and A.P. Moreira. “A systematic literature review on long-term localization and mapping for mobile robots”. In: *Journal of Field Robotics* 40.5 (2023), pp. 1245–1322. DOI: 10.1002/rob.22170.
- [18] R.B. Sousa, H.M. Sobreira, M.F. Silva, and A.P. Moreira. “Line fitting-based corner-like detector for 2D laser scanners data”. In: *2024 10th International Conference on Automation, Robotics and Applications (ICARA)*. Athens, Greece, 2024, pp. 210–216. DOI: 10.1109/ICARA60736.2024.10553006.
- [19] R.B. Sousa, H.M. Sobreira, J.G. Martins, P.G. Costa, M.F. Silva, and A.P. Moreira. “Integrating multi-modal perception into ground mobile robots”. In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 104–111. DOI: 10.1109/ICARSC65809.2025.10970176.

1.6.2 Complementary contributions

- [20] J.D. Ribeiro, R.B. Sousa, J.G. Martins, A.S. Aguiar, F.N. dos Santos, and H.M. Sobreira. “Indoor Benchmark of 3D LiDAR SLAM at iilab – industry and innovation laboratory”. In: *IEEE Access* (2025), pp. 1–23.
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1.6.3 Public datasets

- [22] J.D. Ribeiro, R.B. Sousa, J.G. Martins, A.S. Aguiar, F.N. dos Santos, and H.M. Sobreira. *IILABS 3D: iilab indoor LiDAR-based SLAM dataset*. iilab – industry and innovation lab: INESC TEC – Institute for Systems, Computer Engineering, Technology, and Science, 2025. DOI: 10.25747/VHNJ-WM80.

1.6.4 Open-source Software

- https://github.com/INESCTEC/inesctec_mrdt_line_fit_based_corner_detector_2d_laser: C++ implementation compatible with Robot Operating System (ROS) 1 of the line fitting-based corner-like detector for 2D lasers [18];
- https://github.com/INESCTEC/inesctec_mrdt_hangfa_discovery_q2: repository containing all the documentation (hardware, 3D modelling, electronics, network configuration, sensor setup) and code (firmware, C++ nodes compatible with both ROS 1 and ROS 2) associated with the modifications made on the Hangfa Discovery Q2 omnidirectional platform [23] to integrate multimodal perception [19];
 - https://oss.inesctec.pt/inesctec_mrdt_hangfa_discovery_q2/: GitHub.io public website to help researchers replicate the multimodal perception integration discussed in [19].
- <https://github.com/sousarbarb/slr-ltIm-mr>: repository containing all the documentation, data, and Python scripts used in the scope of the systematic literature review on long-term localization and mapping [17];
- https://github.com/sousarbarb/hector_slam: added an offline node to the original Hector SLAM [24] open-source repository (https://github.com/tu-darmstadt-ros-pkg/hector_slam) to process multiple ROS bag files, generate the 2D occupancy grid map, and save the robot pose data into TUM format [25];
- https://github.com/sousarbarb/slam_gmapping: same addition made to the `hector_slam` ROS package for the GMapping [26] open-source implementation;
- https://github.com/sousarbarb/stage_ros: native ROS wrapper for the Stage Simulator [27–29], added compatibility with both ROS 1 and ROS 2 versions;
- https://gitlab.com/sousarbarb_srrg-software/srrg_system: added CMake compatibility with both ROS 1 and ROS 2 versions of the packages `srrg2_core`, `srrg2_qgl_viewport`, and `srrg2_solver` [14].

1.7 Document Structure

The remaining of this document is organized as follows ([**TODO: update later the following text**]):

- Chapter 2 presents concepts and methodologies for the SLAM problem, covering scan matching, Extended Kalman Filter (EKF)-based SLAM, particle filters, and pose graph formulations. The chapter

also reviews the literature on long-term SLAM in dynamic environments, including map representation, map matching, dynamic object detection, and dynamics prediction modeling;

- Chapter 3 describes the various mobile platforms and datasets used in this PhD thesis, including robots such as the 5dpo omnidirectional robot, the Avantgarde tricycle and Flowbotics differential autonomous forklifts, and also details the IILABS 3D and other datasets from the literature;
- Chapter 4 presents a corner-like feature detection algorithm for 2D laser-scanner data and evaluates its geometric invariance to various viewpoints;
- Chapter 5 explores the concept of using distance maps for data registration, discussing optimization algorithms, covariance estimation, and point-to-point and point-to-plane alignment error formulations;
- Chapter 6 presents the research conducted in this PhD thesis on SLAM in dynamic environments, focusing on techniques such as online SLAM, dynamics removal, and offline global bundle adjustment;
- Chapter 7 summarizes the conclusions and contributions of this research and offers final remarks on future work, including potential areas for further exploration and involvement in the scientific community.

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Chapter 2

Literature Review

TBC

2.1 Simultaneous Localization and Mapping (SLAM)

TBC

2.1.1 Problem introduction

2.1.2 Scan matching

2.1.3 Extended Kalman Filter (EKF) SLAM

2.1.4 Particle filter

2.1.5 Pose graph

2.2 Systematic Literature Review

TBC

2.2.1 Review methodology

2.2.2 Results overview

2.3 Long-Term SLAM in Dynamic Environments

TBC

2.3.1 Map representation**2.3.2 Map matching****2.3.3 Dynamic objects detection****2.3.4 Prediction modeling****2.4 Summary and Conclusions**

TBC

Chapter 3

Datasets

TBC

3.1 Mobile Platforms

TBC

3.1.1 5dpo three-wheeled omnidirectional robot

3.1.2 Modified Hangfa Discovery Q2 four-wheeled omnidirectional platform

3.1.3 IDEPA's tricycle autonomous robot

3.1.4 Avantgarde tricycle autonomous forklift

3.2 Faculdade de Engenharia da Universidade do Porto

TBC

3.3 IILABS 3D – iilab Indoor LiDAR-based SLAM

TBC

IDEPA

Cappero

Avantgarde at COLEP

Kuka

3.4 Summary and Conclusions

TBC

Chapter 4

Line Fitting-Based Corner-Like Detector for 2D Laser Scanners Data

TBC

4.1 Background

TBC (background / literature analysis specific to the topic of this chapter)

4.1.1 Line fitting intersection

4.1.2 Curvature-based detection

4.2 Corner-Like Feature Detector

TBC

4.2.1 Linear least-squares regression

4.2.2 Feature extraction

4.3 Experimental Results and Discussion

TBC

4.3.1 Feature extraction results**4.3.2 Features observability****4.3.3 Features association distance****4.3.4 Integration with a line features extractor algorithm****4.4 Summary and Conclusions**

TBC

Chapter 5

Data Registration with Distance Maps

TBC

5.1 Background

TBC (ICP and its variants, etc.)

5.2 Distance Maps

TBC (unsigned distance map structure, types of distance map formulations – only NN, NN + distance, NN + 1st derivative, NN + 1st and 2nd derivative).

5.3 Data Registration

TBC

5.3.1 Least-squares on manifolds

TBC

5.3.2 Optimization algorithm

5.3.3 Covariance estimation

5.3.4 Problem formulation – point-to-point error formulation

TBC

5.3.4.1 State space**5.3.4.2 Measurement space****5.3.4.3 Prediction function****5.3.4.4 Error function****5.3.5 Problem formulation – point-to-plane error formulation**

TBC

5.3.5.1 State space**5.3.5.2 Measurement space****5.3.5.3 Prediction function****5.3.5.4 Error function****5.4 Experimental Results and Discussion**

TBC

5.4.1 Alignment results**5.4.2 Covariance estimation****5.4.3 Computation performance****5.5 Summary and Conclusions**

TBC

Chapter 6

Simultaneous Localization and Mapping (SLAM) in Dynamic Environments

TBC

6.1 Background

TBC

6.2 Online SLAM

TBC

6.3 Dynamics Removal

TBC

6.4 Offline Global Bundle Adjustment

TBC

6.5 Experimental Results and Discussion

TBC

6.6 Summary and Conclusions

TBC

Chapter 7

Conclusions and Future Work

TBC

7.1 Conclusions and Contributions

TBC

7.2 Final Remarks and Future Work

7.2.1 Future Work

7.2.2 Involvement in the Scientific Community

TBC

7.2.2.1 Collaborations

Giorgio Sapienza, Rome (PhD Internship, 6 months, February - July 2024).

5dpo Robotics Team:

- [30] R.B. Sousa, P.G. Costa, and A.P. Moreira. “A pose control algorithm for omnidirectional robots”. In: *2021 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Santa Maria da Feira, Portugal, 2021, pp. 91–96. DOI: 10.1109/ICARSC52212.2021.9429803
- [31] R.B. Sousa, C.D. Rocha, J.G. Martins, J.P. Costa, J.T. Padrão, J.M. Sarmiento, J.P. Carvalho, M.S. Lopes, P.G. Costa, and A.P. Moreira. “A robotic framework for the robot@factory 4.0 competition”. In: *2024 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Paredes de Coura, Portugal, 2024, pp. 66–73. DOI: 10.1109/ICARSC61747.2024.10535935
- [32] M.S. Lopes, J.D. Ribeiro, A.P. Moreira, C.D. Rocha, J.G. Martins, J.M. Sarmiento, J.P. Carvalho, P.G. Costa, and R.B. Sousa. “From competition to classroom: a hands-on approach to robotics learning”. In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 170–176. DOI: 10.1109/ICARSC65809.2025.10970153

7.2.2.2 Scientific projects

APPLICATION: SLAM4all (SLAM em tempo real para multi-domínio com múltiplos robôs sob transmissão de dados limitada).

PROJECTS: VINCI7D (Gerber parser); GreenAuto; PRODUTECH R3.

- PRODUTECH R3:

- [33] T.B. Levin, J.M. Oliveira, R.B. Sousa, M.F. Silva, B.S. Parreira, H.M. Sobreira, and H.S. Mendonça. “Image and command transmission over the 5G network for teleoperation of mobile robots”. In: *2024 7th Iberian Robotics Conference (ROBOT)*. Madrid, Spain, 2024, pp. 1–8. DOI: 10.1109/ROBOT61475.2024.10797434
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- [35] F.D. Pacheco, P.M. Rebelo, R.B. Sousa, M.F. Silva, and H.S. Mendonça. “Integrated RFID system for intralogistics operations with industrial mobile robots”. In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 124–131. DOI: 10.1109/ICARSC65809.2025.10970172

- VINCI 7D:

- [36] R.B. Sousa, C. Rocha, H.S. Mendonça, A.P. Moreira, and M.F. Silva. “Gerber file parsing for conversion to bitmap image—the VINCI7D case study”. In: *IEEE Access* 10 (2022), pp. 69659–69679. DOI: 10.1109/ACCESS.2022.3187042

7.2.2.3 Supervision

B.Sc. supervision (not official) at FEUP (February - June 2023; 2nd semester 2022/23): Rodrigo de Vasconcelos e Miguel (Compensação do Movimento do Robô nos Dados de Sensores Exteroceptivos); José Miguel Guedes (Controlo de Trajetórias de Robôs Móveis com o Hangfa Discovery Q2); André Luís Pires (Robótica e Sistemas Autónomos em Localização e Mapeamento de Robôs Móveis com o 3-omni 5DPO MSL); Ana Luísa Mota (Robótica e Sistemas Autónomos em Localização e Mapeamento de Robôs Móveis com o Hangfa Discovery Q2).

B.Sc. supervision (not official) at ISEP (March - July 2023; 2nd semester 2022/23): Paulo Ricardo Barbosa (Teste de algoritmos de mapeamento open source recorrendo ao sensor RS-Helios-5515).

BII at INESC TEC: Daniele da Mota Caldana (Aplicação para deteção de local para manipulação de paletes e paletes danificadas); Duarte Filipe Monteiro (Caixas metálicas GreenAuto); David José Castro (Caixas metálicas GreenAuto); João Miguel Oliveira (tele-controlo AMRs); Thiago Baldassarri Levin (comunicações 5G).

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